From variant scores to molecular mechanisms

Interpreting human genetic variation at atomic resolution

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The contribution: make molecular context explicit

A Perspective, rather than a new prediction benchmark
Computational structural genomics (CSG) combines sequence annotations, structures, dynamics and cellular evidence to generate mechanistic hypotheses about variants.

The proposed "avatar" framework organizes evidence at gene, protein and complex scales. It is an organizing concept; a complete multiscale software implementation remains to be developed.

The paper's evidentiary base comprises prior applications, a literature-adoption survey, and a formal distinction between molecular damage and clinical pathogenicity.

Main text, introduction and avatar-framework section; Boxes 1–4; Supplementary Notes 1–2.

The scientific question: what does a variant actually do?

A sequencing result identifies a change. Interpretation asks how that change alters a biological system and whether that alteration explains disease.

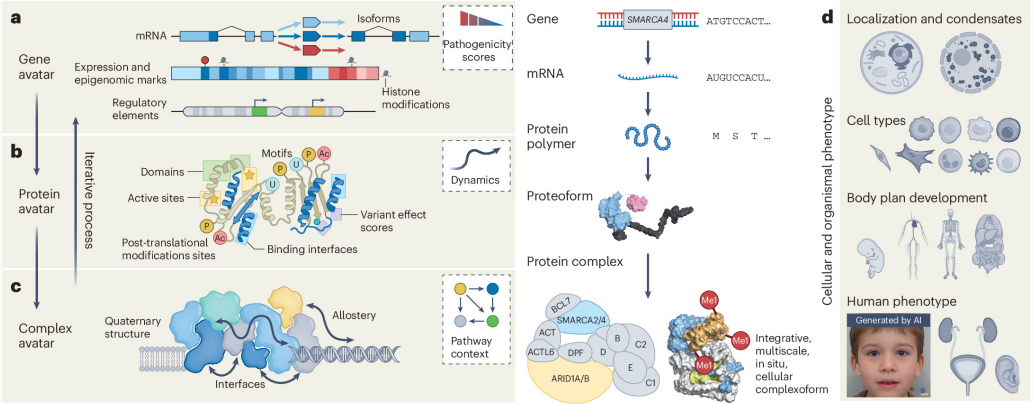
Molecular question
Does the change affect folding, binding, catalysis, dynamics or assembly?

Individual-level question
Does that effect matter in this person's genetic background, exposures and clinical context?

These are connected questions with different evidence requirements.

Abstract; Supplementary Note 2. Paraphrased conceptual argument.

Figure 1: interpretation crosses biological scales



Zimmermann & Urrutia, Fig. 1. Original publisher figure; conceptual integration, not benchmark data.

An “avatar” is an organized representation of context

Layer	What the representation contributes
Gene	Isoforms, conservation, expression, regulation and disease associations.
Protein	Domains, motifs, modifications, structure, dynamics and binding sites.
Complex	Interfaces, composition, allosteric coupling, localization and pathway context.

Arrows between layers matter: a residue is interpreted in the protein, and that protein in its functional assembly and cellular setting.

Fig. 1a–d caption. “Avatar” is the authors’ representational shorthand.

Running biological example: SMARCA4 is more than a residue

Gene: patient sequencing identifies a variant in SMARCA4, a gene implicated in Coffin–Siris syndrome.

Protein: the affected position could map to a helicase, bromodomain or interaction domain, suggesting different molecular questions.

Complex and cell: SMARCA4 acts within BAF chromatin-remodeling complexes; composition and chromatin environment can vary across cell types and development.

Interpretation
A confident coordinate model does not by itself specify which function, assembly or developmental context is disrupted.

Fig. 1 caption: SMARCA4 domain structures and cBAF structure are illustrative; no new patient assay is reported there.

A practical workflow suggested by the framework

1. Map the variant to the relevant transcript, protein region and molecular assembly.
2. Ask a specific mechanism question: stability, an interface, a catalytic site or allosteric coupling?
3. Compare reference and variant in appropriate structural states; use dynamics when a static model is insufficient.
4. Test the prediction with an assay matched to that mechanism; integrate individual-level evidence separately.

Output: a testable mechanism with uncertainty, rather than just a scalar score.

Teaching synthesis of Boxes 1–3 and implementation discussion; not a reported software pipeline.

Toy example: the same substitution can imply different tests

Location	Candidate mechanism	Discriminating experiment
Buried protein core	Destabilized fold	Abundance and stability
Partner interface	Weaker assembly	Binding or complex formation
Catalytic region	Altered chemistry	Activity with abundance control

Suppose a hydrophobic residue becomes charged. Its chemical change alone does not select one of these mechanisms; structural context helps prioritize the test.

Hypothetical teaching example, not a variant or measured result from this Perspective.

Why include dynamics and multiple states?

A protein is represented by a distribution of conformations, not only a single coordinate set.

A variant could preserve the visible fold yet alter the relative occupancy of functional states or the coupling between distant sites.

What a simulation comparison needs
Comparable reference/variant conditions, a biologically relevant assembly, replication, and an observable linked to the proposed mechanism.

A changed trajectory is evidence about a model. Experimental validation and sampling uncertainty still determine its biological interpretation.

Box 2: reference/variant calculations, conditions and positive/negative controls; replication is journal-club interpretation.

Formalize the two questions before combining them

D	Molecular function is damaged.
P	The variant is pathogenic.
S, C, M	Affected site, molecular context and perturbed mechanism.
G, E	Individual genetic background and environmental exposures.

$$\underbrace{\Pr(D \mid S, C, M)}_{\text{molecular evidence}} \qquad \underbrace{\Pr(P \mid D, G, E)}_{\text{individual/disease evidence}}$$

"Damaging" and "pathogenic" are different events.

Supplementary Note 2. Schematic formalization; not a calibrated clinical calculator.

The product formula requires explicit assumptions

Let $X = (S, C, M, G, E)$. The law of total probability gives

$$\Pr(P \mid X) = \Pr(P \mid D, X)\Pr(D \mid X) + \Pr(P \mid \neg D, X)\Pr(\neg D \mid X).$$

The simplified protein-function model assumes:

1. Pathogenicity outside modeled protein damage is negligible.
2. Given damage and G, E, site/context/mechanism add no further information about pathogenicity.
3. G, E do not materially change the probability of molecular damage.

Then $\Pr(P \mid X) \approx \Pr(P \mid D, G, E)\Pr(D \mid S, C, M)$.

Supplementary Note 2; total-probability derivation written explicitly here to expose the assumptions.

Toy calculation: high molecular confidence need not settle disease

Assume, purely for illustration,

$$\Pr(D \mid S, C, M) = 0.90, \qquad \Pr(P \mid D, G, E) = 0.20.$$

Under the simplified assumptions, $\Pr(P \mid X) = 0.18$.

Increasing molecular confidence to 0.99 gives 0.198; it does not remove uncertainty about whether that damaged function causes disease in context.

Lesson
Better molecular models solve one part of the inference problem. Their output is not automatically an individual-level diagnosis.

Invented numerical teaching example; probabilities are not estimated or validated by the authors.

The broader framework admits mechanisms outside protein damage

Supplementary Note 2 introduces a residual term

$$\mathcal{G} = \Pr(P \wedge \neg D \mid S, C, M, G, E).$$

This captures pathogenic contributions outside the simplified protein-damage route. The authors discuss regulatory effects, RNA stability and other contextual mechanisms.

Discussion: a coding substitution could also alter splicing or expression. A protein-only calculation can miss a relevant route entirely.

The supplement explicitly states that this is a schematic framework, not a replacement for variant-classification standards.

Supplementary Note 2. Coding/splicing example is a teaching illustration.

Empirical evidence here: a literature-adoption survey

Question: how frequently do structure/dynamics methods appear in human-genetics literature?

Supplementary Note 1 reports programmatic PubMed queries executed on 24 April 2026:

1. Search structure/dynamics and human-genetics terms in titles, abstracts and MeSH.
2. Exclude drug-development literature to focus on variant mechanisms.
3. Repeat in a selected panel of medical/genomics journals; compare with control terms.

The unit is a retrieved publication, not a patient, variant or diagnostic decision.

Supplementary Note 1, literature assessment methods and query table.

Main survey result: sparse absolute coverage in selected journals

Selected "Top-42" journal panel	Count	Share
Human-genetics query hits	126,520	100%
Also structure-method hits	208	0.164%
Also molecular-dynamics hits	55	0.043%

The supplement reports roughly 92% overlap with its animal-modeling control.

What follows: structure/dynamics terms are uncommon in this chosen corpus. This does not measure how many variants were resolved or whether CSG improves outcomes.

Supplementary Note 1, Top-42 table. Percentages use 126,520 as denominator.

A low fraction and an enriched overlap can coexist

In the Top-42 panel, only 460 of 860,373 papers match structure terms; 208 are also among the 126,520 genetics hits.

$$\mathrm{OR} = \frac{208 \times 733{,}601}{126{,}312 \times 252} \approx 4.79.$$

So the overlap is enriched relative to independence *within this panel*, despite the small absolute fraction of genetics papers using these terms.

The broader-corpus analysis instead reports depletion. These are different populations and should not be merged into one universal adoption statistic.

Supplementary Note 1, odds-ratio calculations and journal-panel table.

Audit point: survey definitions constrain the headline

The drug-excluded odds-ratio block uses $n = 38{,}645{,}914$, genetics $= 5{,}150{,}022$, intersection $= 3{,}005$, and structural/MD marginal $= 146{,}884$; it reports $\log_2 \mathrm{OR} = -2.885$.

But the query table gives **125,181** for MD after drug exclusion. Using that marginal yields $\log_2 \mathrm{OR} \approx -2.649$.

This is an internal denominator inconsistency, not evidence that the broad sparsity claim reverses.

Other qualifications: the MD query includes structural-homology MeSH; the journal panel is a heuristic selection, not a formal ranking. Retrieval controls do not prove perfect sensitivity or specificity.

Supplementary Note 1. Recalculation from reported counts; no PubMed search rerun.

What the cited applications demonstrate

Application	Evidence summarized by the Perspective
TGFBR2 kinase variants	Modeling and MD generate variant-specific mechanistic hypotheses.
935 RAS-family variants	3D scores correlate with biochemical measurements and group nonhotspot variants with functional classes.
Clinical exome program	Among 1,101 patients, integrated translational research contributed 100 diagnoses; overall yield was 36.5%.

The exome result is a combined research workflow, not a CSG-only effect or an incremental 36.5-point gain.

Main text, example case and obstacles sections; cited refs. 48, 52, 61, 126–129. Summaries of prior studies, not new trials.

Tiered deployment: where the authors expect near-term value

Prioritize unresolved, consequential VUS, disease-relevant genes/complexes, discordant predictions, or cases where a mechanism will determine the next assay.

Technical limits: single ligand-free models can miss relevant states, modifications and partners; MD is computationally and operationally demanding. Disordered and noncoding regions require more assumptions and orthogonal evidence.

Implementation limits: trained analysts, validation standards and reviewable reports are needed. Reports should identify the model, mapping, confidence, settings, controls, intended use and uncertainty.

Main text, "Current obstacles to a broad adoption of CSG"; Boxes 1–3.

What would make the proposal convincing in practice?

1. **Mechanism validation:** blinded variants, reference controls, and assays distinguishing stability, activity and assembly.
2. **Added information:** comparison with sequence-based scores, using held-out genes or families and avoiding evidence reuse.
3. **Clinical utility:** whether adding CSG changes well-supported classifications or directs successful follow-up experiments.

These are proposed evaluation criteria. The Perspective cites prior applications, but does not provide a unified prospective evaluation of this full framework.

Journal-club assessment; distinguish decision utility emphasized by the authors from prediction accuracy alone.

Discussion: promising inference, explicit boundaries

Take-home: structural context can turn a variant into a testable mechanism; disease attribution still requires a separate contextual inference.

1. For a new variant, which mechanism-specific assay would a structure actually change?
2. When do cell type, binding partners or background violate the product formula's assumptions?
3. Is low publication frequency an adoption gap, a measurement artifact, or appropriate selection of cases?

Primary reading: Zimmermann MT & Urrutia R. *Nature Genetics* (2026), doi:10.1038/s41588-026-02763-z. Figure 1; Boxes 1–4; Supplementary Notes 1–2. Teaching examples are explicitly hypothetical.